

Partial Result: Unconditional Support Ideals Are Non-Pathological

Automatic Math Discovery Run `fa_banach_001`

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Status

This packet records a **partial result** related to Question 11.2 of Kwela-Swaczyna, *Zoo of ideal Schauder bases*, arXiv:2508.21235. The full question remains open.

Source Question

The source paper proves in Theorem 7.1 that every non-pathological analytic P -ideal can occur as a critical ideal $\mathcal{CR}(X, \hat{a})$. It then asks:

Recall that by Theorem 7.1, for every non-pathological analytic P -ideal $\mathcal{I}$ there are a Banach space $X \subseteq \mathbb{R}^\omega$ and $\hat{a}$ simple over $\hat{e}$ such that $\mathcal{CR}(X, \hat{a}) = \mathcal{I}$.

Question 11.2. *Let $\mathcal{I}$ be an analytic P -ideals. Are there a Banach space X and $\hat{a}$ such that $\mathcal{CR}(X, \hat{a}) = \mathcal{I}$?*

Thus the remaining difficulty is the possible realization of pathological analytic P -ideals as critical ideals of Banach-space ideal Schauder bases.

Definitions

Let $\phi : \mathcal{P}(\omega) \rightarrow [0, \infty]$ be a lower semicontinuous submeasure. Recall that

$$\text{Exh}(\phi) = \{A \subseteq \omega : \lim_{m \rightarrow \infty} \phi(A \setminus m) = 0\}.$$

Following the source paper, ϕ is *non-pathological* if

$$\phi(A) = \sup\{\mu(A) : \mu \text{ is a finitely supported measure and } \mu(B) \leq \phi(B) \text{ for every } B \subseteq \omega\}$$

for every $A \subseteq \omega$.

Idea of the Proof

The construction in Theorem 7.1 of the source paper realizes a given non-pathological analytic P -ideal by making membership of a set A mean that a positive weighted subseries indexed by A converges in a Banach sequence space. The theorem below proves a converse for this whole

positive-series mechanism. Once a 1-unconditional basis is present, the norm of finite positive sums defines a lower semicontinuous submeasure ϕ . Hahn–Banach, together with unconditionality, supplies positive norming functionals for finite positive sums; these functionals are exactly the finitely supported dominated measures witnessing that ϕ is non-pathological.

Theorem

Theorem 1. *Let X be a Banach space with a 1-unconditional Schauder basis $(u_n)_{n \in \omega}$, and let $c_n > 0$ for all $n \in \omega$. Define*

$$\mathcal{J}_{X,c} = \left\{ A \subseteq \omega : \sum_{n \in A} c_n u_n \text{ converges in } X \right\}.$$

Then there is a non-pathological lower semicontinuous submeasure ϕ on ω such that

$$\mathcal{J}_{X,c} = \text{Exh}(\phi).$$

One may take

$$\phi(A) = \sup \left\{ \left\| \sum_{n \in F} c_n u_n \right\| : F \in [A]^{<\omega} \right\}.$$

Proof. The displayed formula defines a submeasure: monotonicity is immediate, and subadditivity follows from the triangle inequality by splitting each finite $F \subseteq A \cup B$ into $F \cap A$ and $F \cap B$. It is lower semicontinuous because it is the supremum over finite subsets:

$$\phi(A) = \sup_m \phi(A \cap m).$$

We first identify $\mathcal{J}_{X,c}$ with $\text{Exh}(\phi)$. Suppose $\lim_m \phi(A \setminus m) = 0$. Then finite tails of $\sum_{n \in A} c_n u_n$ have arbitrarily small norm, so the net of finite partial sums is Cauchy and the subseries converges in X .

Conversely, suppose $\sum_{n \in A} c_n u_n$ converges, and let

$$z_m = \sum_{\substack{n \in A \\ n \geq m}} c_n u_n$$

be its m -th tail. Since (u_n) is 1-unconditional, the induced lattice norm is monotone on positive vectors: whenever $0 \leq y \leq z$ coordinatewise, $\|y\| \leq \|z\|$. Hence for every finite $F \subseteq A \setminus m$,

$$\left\| \sum_{n \in F} c_n u_n \right\| \leq \|z_m\|.$$

Taking the supremum over such F gives $\phi(A \setminus m) \leq \|z_m\|$, and $\|z_m\| \rightarrow 0$. Therefore $A \in \text{Exh}(\phi)$.

It remains to prove that ϕ is non-pathological. Fix a finite $F \subseteq \omega$, and put

$$x_F = \sum_{n \in F} c_n u_n.$$

By Hahn–Banach, there is $f \in S_{X^*}$ such that $f(x_F) = \|x_F\|$. Since the basis is 1-unconditional, replacing f by its coordinatewise absolute value gives a positive functional $g \in S_{X^*}$ with

$$g(x_F) = \|x_F\| = \phi(F).$$

Define the finitely supported measure

$$\mu_F(\{n\}) = \begin{cases} c_n g(u_n), & n \in F, \\ 0, & n \notin F. \end{cases}$$

For every $B \subseteq \omega$,

$$\mu_F(B) = g\left(\sum_{n \in B \cap F} c_n u_n\right) \leq \left\| \sum_{n \in B \cap F} c_n u_n \right\| \leq \phi(B),$$

while $\mu_F(F) = \phi(F)$. Thus each finite value $\phi(F)$ is attained by a finitely supported measure dominated by ϕ . Since ϕ is lower semicontinuous,

$$\phi(A) = \sup\{\phi(F) : F \in [A]^{<\omega}\}$$

for all $A \subseteq \omega$, and the preceding measures show exactly that ϕ is non-pathological. $\square$

Corollary 1. *The positive weighted-subseries construction used in Theorem 7.1 of arXiv:2508.21235 cannot be extended to realize pathological analytic P -ideals merely by changing the Banach sequence norm while retaining a 1-unconditional basis.*

Proof. Theorem 7.1 realizes a non-pathological analytic P -ideal $\mathcal{I}$ by arranging that $A \in \mathcal{I}$ exactly when a weighted positive subseries $\sum_{n \in A} c_n u_n$ converges in a Banach sequence space with an unconditional basis. The theorem above shows that every ideal obtained in this way is $\text{Exh}(\phi)$ for a non-pathological lsc submeasure ϕ . Thus a pathological analytic P -ideal cannot be obtained by this route. $\square$

Limitations

This does not answer Question 11.2 in full. It rules out one natural route: realizing the missing pathological analytic P -ideals by the same unconditional positive-series mechanism used in Theorem 7.1. It does not exclude more general ideal bases, non-unconditional constructions, or constructions whose critical ideals are not simply support ideals of positive weighted subseries.

Verification and Novelty Check

The proof is purely abstract; no finite computation is used. The only code in the packet is the reproducible crop script for the source question.

Local registry, solution, attempt, and proof-gap indexes had no exact prior packet for arXiv:2508.21235 or this obstruction. A bounded web search for the exact title and phrases “ideal Schauder basis”, “critical ideal”, and “analytic P -ideal” found no later answer to Question 11.2. The source paper itself proves the forward non-pathological realization theorem, but not the converse obstruction for unconditional support ideals recorded here.

Human Review Recommendation

Review as a likely valid partial result. The main checks are the use of a positive norming functional for positive finite sums in a 1-unconditional basis and the equivalence between convergence of the positive subseries and the tail condition $\lim_m \phi(A \setminus m) = 0$.

References

- [1] A. Kwela and J. Swaczyna, *Zoo of ideal Schauder bases*, arXiv:2508.21235.
- [2] S. Solecki, *Analytic ideals and their applications*, Ann. Pure Appl. Logic 99 (1999), 51–72.